A ring search for the megarena's carriers

The library's global pipeline with its strongest-bin peak search and with a ring search, on fmac renders of megarenas

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Abstract

The megarena detector starts by finding the pattern's two carriers in the spectrum of the whole picture. Today `PatternPhase::peaksSearch` takes the **strongest bin** of the spectrum, then the strongest bin in another direction. This note adds a second search, the **ring search**, which judges every bin against the bins at the same distance from the centre, and compares the two inside the library's own global pipeline: one FFT, one Gaussian band-pass per carrier, phase planes, then the megarena code. Nothing else changes. Four megarenas are rendered by fmac, the Vernier project's marker renderer, at 44 poses, and each picture is spoiled four ways: 704 pictures. On each, both searches run; the carriers they demodulate are compared with the true ones, and the decoded position with the true one.

Key findings

- **The ring search finds the right carriers twice as often**: 414 of 704 pictures against 196. On every picture where the strongest bin found them, the ring search found them too.
- **The decoded position gains less**: right on 155 pictures against 99, 1.6 times as often. The global pipeline itself is now the limit.
- **Square on, the two are level.** Both find the right carriers on all 24 clean square-on views, and over the four conditions both decode 64 of 96 square-on pictures at the right position.
- **At 15° of tilt the ring search doubles the right positions** (44 against 22 of 96 pictures); the strongest bin often takes the megarena code's own line, at a third of the carrier, for a carrier.
- **From 30° the global pipeline fails whatever the carriers.** The ring search still finds the right pair on 60 of 96 pictures at 30°, but only 6 decode at the right position, and none at 50°: a single band-pass cannot follow a carrier whose frequency changes across the picture.
- **The detector never says it failed.** `patternFound()` was true on all 1 408 runs, including the 1 154 decoded at a wrong position, 1 135 of them three periods or more away.

In short: the ring search is a safe change for the global pipeline. It never loses the carriers the strongest bin finds, and decodes more pictures. But on its own it does not make the global pipeline work on tilted views: that needs a measurement that follows the carriers across the picture, and a decoder that can tell a wrong code from a right one.

1 The two searches

The strongest bin (`PatternPhase::peaksSearch`, unchanged). Take the magnitude of the picture's spectrum. Keep the bins between 20 and 500 from the centre, blur them over 3×3 and keep half the plane. The strongest bin is the first carrier. Cut a thin wedge around its direction (its width set by the band-pass $\sigma = 3$) and the opposite one; the strongest bin left is the second carrier.

The ring search (`PatternPhase::ringPeaksSearch`, new). In plain words:

- **It fades the picture to zero at its edges** (a Hann window, after taking the mean out) before its own FFT, so the picture's border does not draw a bright cross through the spectrum.
- **It sums each bin with its 8 neighbours**, so a carrier spread over a few bins still counts in full.
- **It scores each bin against its ring**: its value over the mean of all bins at the same distance from the centre, one bin wide. Edges, shadows and clutter put their energy at low frequencies, but evenly around the ring, so they score low; a carrier is a narrow spike and scores high. A ring quieter than the average one is held to the average.

- **It keeps every local maximum that scores at least 0.2 of the best**, from 6 bins out (counted along the picture's short side) to 0.35 cycles per pixel, strongest first.
- **It knows the code's line.** If the spectrum at three times a peak is more than half as strong as the peak, the peak is the megarena code's line at a third of the carrier, and the carrier is taken at three times it.
- **It looks for the second carrier where it can be**: 0.7 to 1.4 times the first one's distance from the centre and at least 60° away.

The first peak that has such a partner gives the pair. Both are put back in the upper half plane and ordered as `peaksSearch` orders them, so the rest of the pipeline is the same. It is switched on with `detector.setBool("ringPeakSearch", true)`; the default stays the **strongest bin**. It runs on the CPU path only; the CUDA engine keeps its own search. Figure 1 shows both on the same board.

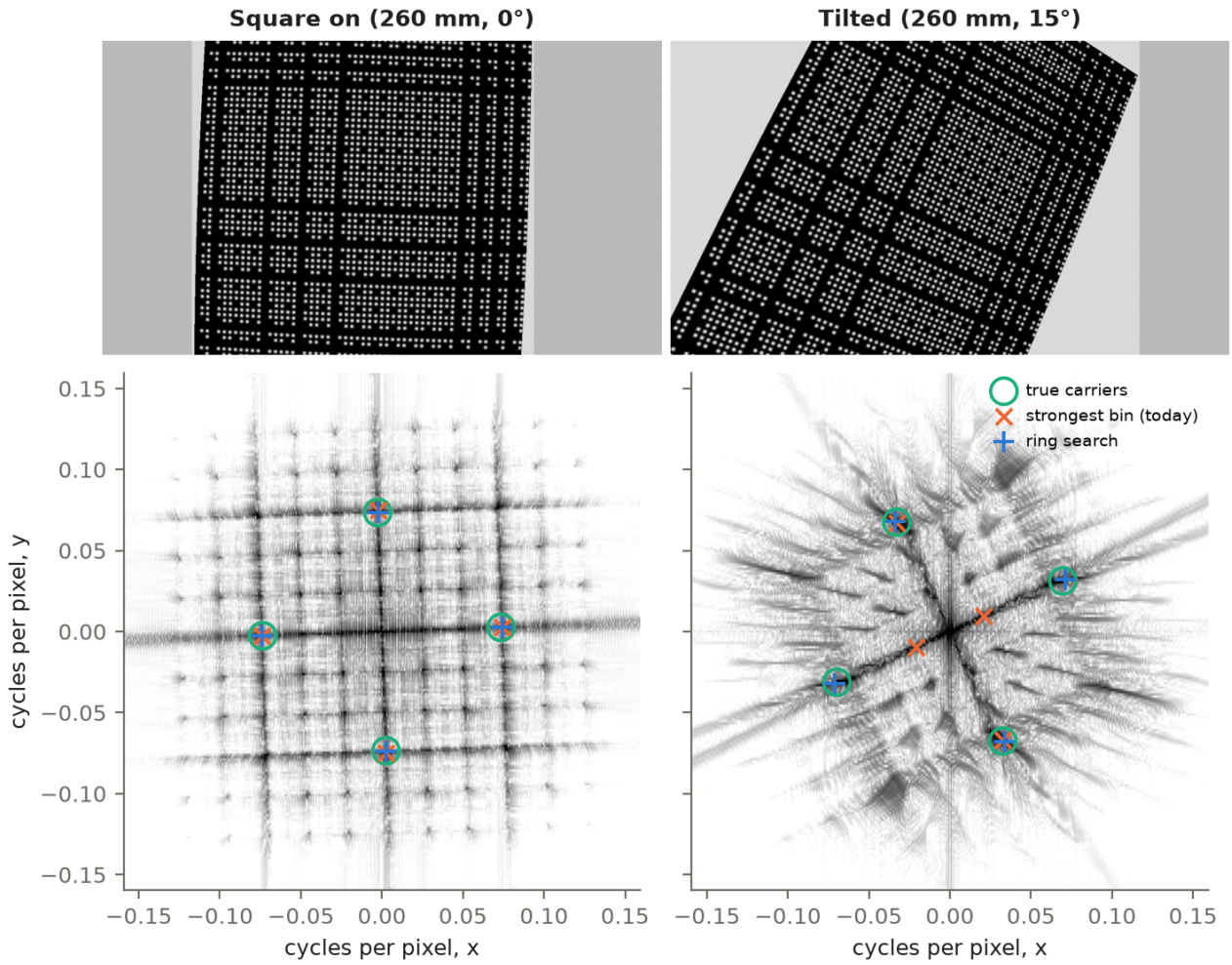


Figure 1: The 3.54 mm megarena at 260 mm, square on (left) and tilted by 15° (right), and the middle of each picture's spectrum, with the bins each search picked. Square on, both pick the true carriers. Tilted, the **strongest bin** takes the code's line at a third of one carrier, close to the centre; the **ring search** stays on the carriers.

2 The setup

The pictures. fmac renders through a thin-lens camera: a 6 mm lens at f/2 on 6 µm pixels, focused at 300 mm, 1280×720, $f_x = f_y = 1000$ px, with lens distortion, diffraction and the Rec. 709 transfer curve, 8-bit output, undone back to linear intensities. The four megarenas are 280×200 mm, with dot pitches of 2.5, 3.54, 5 and 7.07 mm and a code of order 6. Each is rendered at 44 poses: 20 tilted 20–40° and placed around the frame, 260–380 mm off; and 24 test views at 220 to 450 mm, square on and tilted by 15, 30 and 50°.

What is added. Each picture is used four ways: as rendered; with Gaussian noise of $\sigma = 0.03$; with three hard-edged *shadows*, rectangles of a fifth to three fifths of the frame darkened to 40–80 %; and with a *blind*, a strong stripe pattern of 8–24 px period at a random angle covering a third of the frame's width. That makes 704 pictures.

What is counted. The detector is `MegarenaPatternDetector` with the pitch as its period and code size 6, everything else at its defaults. Per picture and search:

- *Right carrier pair*: the gradients of the two fitted phase planes, the carriers actually demodulated, both lie within 20 % of the true carriers. The true carriers come from the true pose, at the board point seen at the middle of the picture.
- *Right position*: the decoded (x, y) lies within half a period of the true one on both axes. The detector reports minus the board point under the picture's centre, 15 periods off on both axes, modulo the code's length of 189 periods; that relation was read off the clean square-on views and holds to a hundredth of a period on every board. A misread code lands a whole code bit or more away: of all 1 408 decodes, 254 are within half a period, 19 between half a period and three, and 1 135 three periods or more away.

3 Results

3.1 Overall

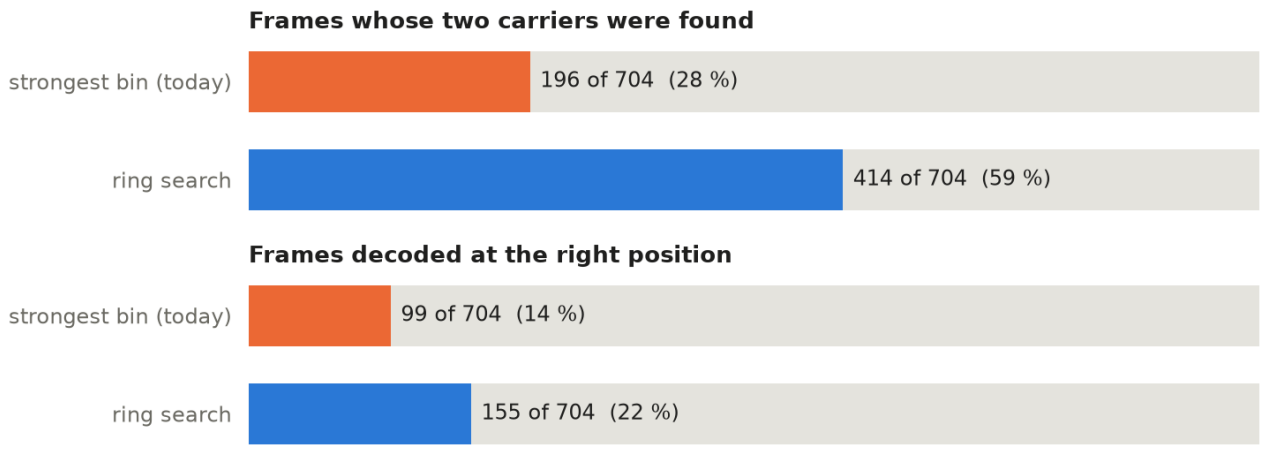


Figure 2: Every picture: 4 boards, 44 poses, 4 conditions. The grey track is all 704.

Figure 2. The **ring search** finds the right carriers on 414 pictures against 196, and never misses a pair the **strongest bin** found. The right position follows less: 155 pictures against 99. The **ring search** decodes 69 pictures the **strongest bin** does not, and loses 13 it does.

Those 13 are not wrong carriers: on 12 of them both searches found the right pair. The two searches sometimes place the same carrier one bin apart, the **ring search** on the maximum of its 3×3 sum, the **strongest bin** on its lightly blurred maximum. The band-pass then sits one bin apart, the phase maps differ slightly, and the code reading, which samples the dots through those maps, can flip: on one clean view at 400 mm the decoded y moved by 108 periods for a one-bin shift of the other carrier. Over the 196 pictures where both found the right pair, this goes the **ring search**'s way: right position with both on 86, with the **ring search** only on 32, with the **strongest bin** only on 12.

3.2 Square on against tilted

Figure 3. Square on, both searches find the right carriers on every view and the right position on nearly every one (23 and 21; over the four conditions, 64 and 64 of 96). At 15° the **strongest bin** finds the right pair on 15 views of 24, the **ring search** on all 24, and the right positions follow: 7 against 16 (22 against 44 of 96 over the four conditions). At 30° the **ring search** still finds 20 right pairs, but only 2 views decode at the right position; at 50°, none with either search. The calibration views, all tilted 20–40°, say the same: right pairs 60 against 189 of 320, right positions 11 against 41.

3.3 Noise, shadows and a blind

Figure 4. Noise and shadows change almost nothing for either search. The blind's stripes are the strongest thing in the picture and both searches take them for a carrier on most of its pictures: 5 and 14 right pairs of 176, 2 and 3 right positions. Getting past it would take trying more than one pair, which needs a decoder that says when a code is wrong.

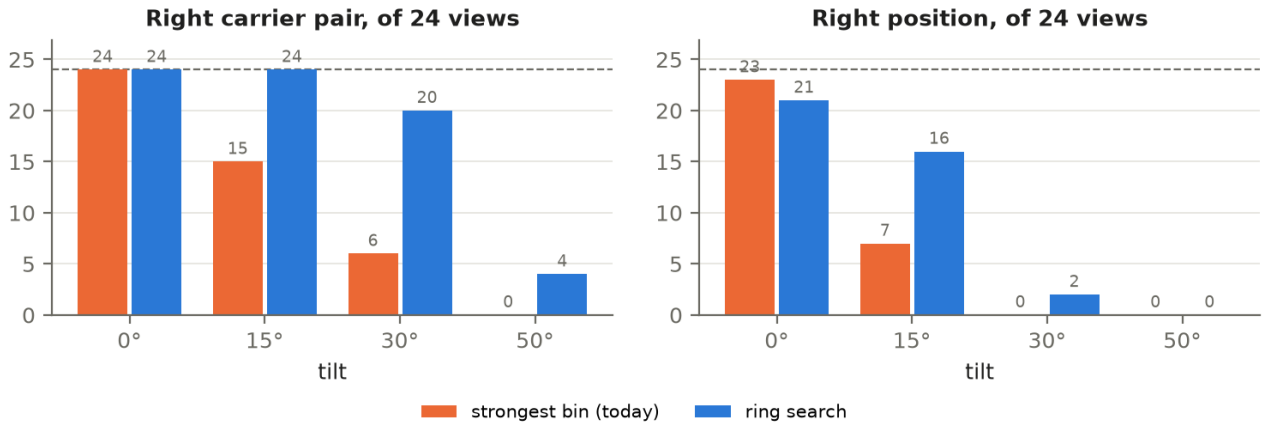


Figure 3: The 24 clean test views by tilt: four boards, six distances from 220 to 450 mm at each tilt. The dashed line is every view.

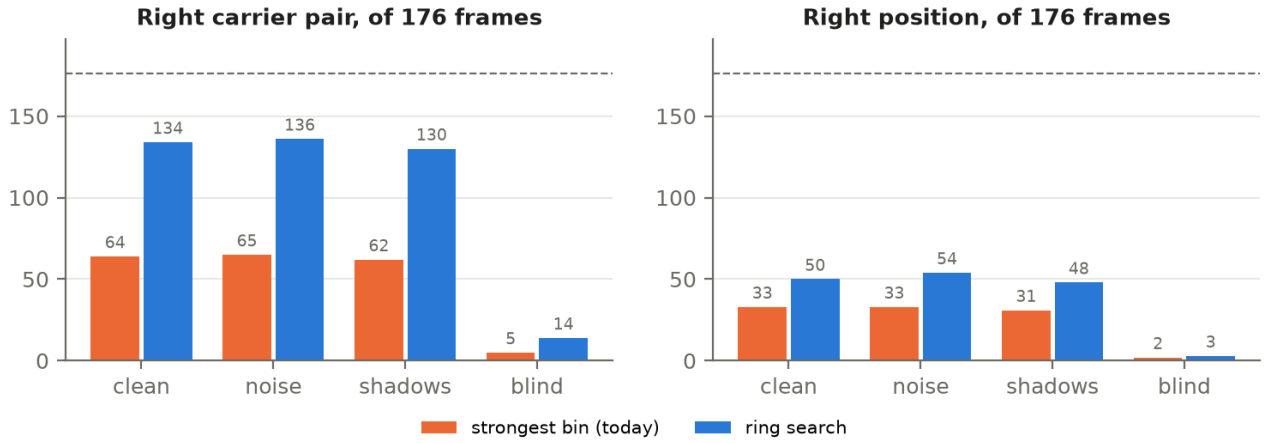


Figure 4: Every picture, by what was added: as rendered, Gaussian noise of $\sigma = 0.03$, three hard-edged shadows, a blind over a third of the frame.

4 Why the gain is smaller in the global pipeline

One band-pass for the whole picture. The pipeline isolates each carrier with one Gaussian band-pass of $\sigma = 3$ bins and fits one phase plane to the whole picture. Square on, the carrier has one frequency everywhere and this is exact. Tilted, the dot spacing changes across the picture, the carrier spreads over more bins than the band-pass keeps, and the phase is no longer a plane. The **ring search** finds the carriers on 259 pictures where the position is still wrong, against 98 for the **strongest bin**; all but 12 of the 259 are tilted views.

No failure signal. `findCodePosition` returns the best match of the sequence read, whether or not it truly matches, and `patternFound()` only checks that two peaks were found. So a misread code comes out as a confident position far from the truth, and the detector cannot fall back to its next candidate pair. Both would help any peak search.

5 What to take from this

- The **ring search** is a safe improvement to the carrier search: square on it changes nothing on average, and elsewhere it finds the carriers twice as often and never loses a pair the **strongest bin** finds. It costs one more FFT of the picture.
- On its own it does not make tilted views work in the global pipeline. The next steps are a measurement that follows the carriers across the picture, and a code check that rejects a sequence not found in the code, so that a wrong decode is reported, and the next candidate pair tried.
- The code reading is sensitive to where the band-pass is centred: one bin can change the decoded position by many periods. Worth a look on its own.
- Caveats: one renderer, one camera, one board size, fixed poses; the noise, shadows and blind were added to the renders, not photographed. The CUDA backend was not changed.

Appendix: reproducing

The comparison is the `comparingPeakSearch` example, which runs both searches of `MegarenaPatternDetector` on every picture listed in a CSV file. `truth.csv` lists the 704 pictures with their pitch, the true board point under the picture's centre and the true carriers; the pictures are 16-bit PNG with linear intensities. With a build of the library and its examples, run from the repository's root:

```
doc/ring-peak-search/run.sh <build-dir> <picture-dir>
```

It writes `results.csv`, one row per picture and search with the pose, the phase planes and the peaks; scores it against the truth with `analyse.py` into `scored.csv`; draws the figures with `plot.py` and builds this PDF. About 11 minutes on 12 cores.